CHAPTER 10

IPv4 Header, Fragmentation & Forwarding

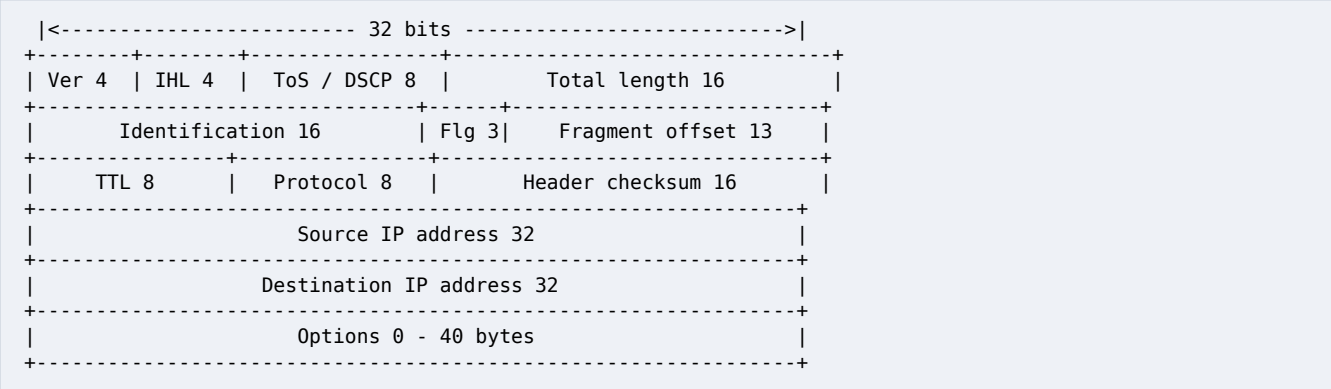
Header fields · MTU · fragment offset · reassembly · TTL · longest-prefix match

QUICK TRICK

Data per fragment = **⌊(MTU − header) / 8⌋ × 8**. Offset = (bytes before it) ÷ **8**. Every fragment except the last has **MF = 1**. Only the **destination** reassembles. Forward using the **longest matching prefix**.

10.1 IPv4 header

● **LEVEL 2 · EXAM**



Field	Facts
IHL	Header length in **4-byte words**. Min 5 (20 B), max 15 (60 B).
Total length	Header + data, in **bytes**. Max 65535, so max data = 65535 − 20 = 65515 B.
Identification	Same value in all fragments of one datagram.
Flags	3 bits: reserved, **DF** (don't fragment), **MF** (more fragments).
Fragment offset	13 bits, in **units of 8 bytes**.
TTL	Decremented at each router; at 0 the packet is dropped.
Protocol	Upper layer: TCP = 6, UDP = 17, ICMP = 1.
Header checksum	Covers the **header only**; recomputed at every router (TTL changes).

10.2 Fragmentation

● **LEVEL 2 · EXAM**

The problem. Each link has a maximum payload, the **MTU** (Ethernet 1500 B). A bigger datagram must be cut up. **The rules.** (1) Each fragment gets its own IP header (copy of the original, with new total length, offset and MF). (2) Data in every fragment **except the last** must be a **multiple of 8 bytes**. (3) Routers may fragment again. (4) **Only the destination** reassembles. (5) If DF = 1 and the packet is too big, it is **dropped** (ICMP error sent).

Worked example. A datagram of total length 4000 B (header 20, data 3980) meets MTU 1500. Data per fragment = ⌊1480/8⌋ × 8 = **1480**.

Fragment	Data bytes	Total length	Offset	MF
1	1480	1500	0	1
2	1480	1500	185 (1480/8)	1

Fragment	Data bytes	Total length	Offset	MF
3	3980 − 2960 = 1020	1040	370 (2960/8)	0

Fragmenting again. Fragment 2 (1480 B data, offset 185, MF = 1) now meets MTU 576. Data per piece = ⌊556/8⌋ × 8 = **552**.

Piece	Data	Total length	Offset	MF
2a	552	572	185	1
2b	552	572	185 + 69 = 254	1
2c	1480 − 1104 = 376	396	254 + 69 = 323	**1** (original MF was 1)

QUICK TRICK

Number of fragments = ⌈ data ÷ (⌊(MTU − 20)/8⌋ × 8) ⌉. After fragmenting, total bytes on the wire = data + 20 × (number of fragments).

10.3 TTL and the checksum

● LEVEL 1 · BEGINNER

A router subtracts 1 from TTL, drops the packet if it becomes 0 (and sends an ICMP "time exceeded"), and recomputes the header checksum. This limits looping packets and is how *traceroute* works.

10.4 Forwarding: longest-prefix match

● LEVEL 2 · EXAM

A router table lists prefixes. For a destination address, **every** matching entry is a candidate; the router picks the one with the **longest prefix** (most specific). **0.0.0.0/0** matches everything and is the default route.

Prefix	Next hop
10.0.0.0/8	R1
10.1.0.0/16	R2
10.1.2.0/24	R3
0.0.0.0/0	R4 (default)

Destination	Matches	Chosen
10.1.2.77	/8, /16, /24, /0	**R3** (/24)
10.1.9.9	/8, /16, /0	**R2** (/16)
10.200.1.1	/8, /0	**R1** (/8)
192.0.2.1	/0 only	**R4**

Host decision. A host computes (destination AND own mask). If it equals its own network, the destination is on the same LAN (send directly). Otherwise it sends to the default gateway.

10.5 Common mistakes

COMMON MISTAKE

- Using the offset in bytes (it is in **8-byte units**).
- Choosing the first matching entry instead of the longest prefix.
- Giving the last piece of a re-fragmented fragment MF = 0 when the original fragment had MF = 1.
- Counting the header once for all fragments. **Every** fragment carries its own 20 B header.
- Making non-last fragments carry a data size that is not a multiple of 8.

GATE QUICK CHECK — Chapter 10

```
data/frag = floor((MTU-20)/8)*8      offset = bytes_before/8      MF=1 except last
header 20..60 B (IHL 5..15)          total length <= 65535        checksum = header only
TTL-1 per router (0 -> drop)         DF=1 & too big -> drop       reassembly only at dest
Forwarding: longest prefix wins      default route 0.0.0.0/0
```

Practice questions

● **LEVEL 1 · BEGINNER**

Q1. Minimum IPv4 header size and the IHL value for it.

Q2. A header has IHL = 7. How many bytes of options?

Q3. What does a router do when TTL reaches 0?

Q4. Where are fragments reassembled?

● **LEVEL 2 · EXAM**

Q5. A fragment carries data starting at byte 2960 of the original data. Its fragment-offset field?

Q6. Datagram of total length 3000 B (header 20) meets MTU 1000. Fragments, data sizes, offsets and MF?

Q7. Total bytes at the IP layer after the fragmentation in the previous question.

Q8. Longest-prefix match for 10.1.9.9 using the table in 10.4.

● **LEVEL 3 · GATE**

Q9. A last fragment has offset 100, total length 220 and header 20 B (MF = 0). Size of the original data?

Q10. A 1500 B datagram (20 B header, DF = 0) passes MTU 576 then MTU 1500. How many fragments reach the destination?

Q11. A router receives a 1500 B packet with DF = 1 for a link with MTU 1000. What happens?

Q12. Does the IP header checksum change at each router?

Answers

Q	Solution
1	**20 B**, IHL = **5**.
2	Header = 28 B, so options = 28 − 20 = **8 B**.
3	Drops the packet and sends an ICMP time-exceeded message.
4	Only at the **destination host**.
5	2960/8 = **370**.
6	Data 2980 B. Per fragment ⌊980/8⌋×8 = 976. Four fragments: data **976, 976, 976, 52**; offsets **0, 122, 244, 366**; MF **1, 1, 1, 0**.
7	2980 + 4 × 20 = **3060 B**.
8	**R2** (matches /8, /16 and /0; /16 is longest).
9	Its data = 200 B. It starts at 100 × 8 = 800. Original data = 800 + 200 = **1000 B**.
10	Data 1480 → ⌊556/8⌋×8 = 552: pieces 552, 552, 376 = **3 fragments**. The later 1500-MTU link does not merge them.
11	**Dropped**, and an ICMP "fragmentation needed" message is returned.
12	**Yes**: TTL changes, so the checksum is recomputed.

Next: Chapter 11 — Network Address Translation (NAT).